

3D Representation Learning

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July, 7th, 2026

3D representation Learning for Robotics

Autonomous Navigation



Human Robot Interaction



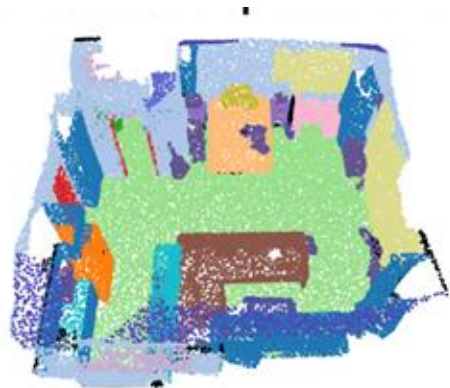
Manipulation and Grasping



3D Scene Understanding



3D representations



Point Cloud

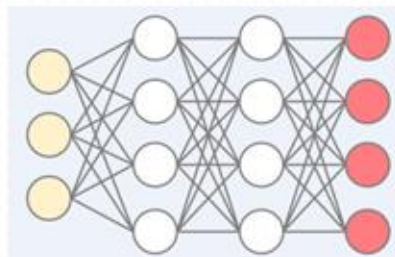


Mesh



Voxels

$$\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^n$$



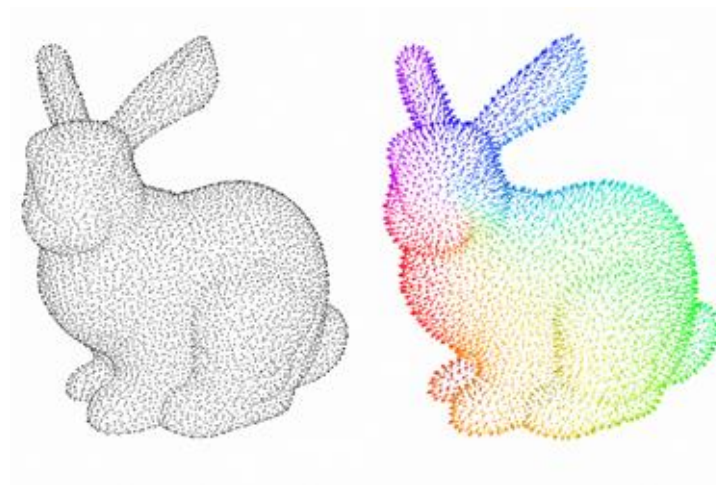
Implicit Representation

E. g. $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^3 | \Phi_{\theta}(\mathbf{x}) = 0\}$

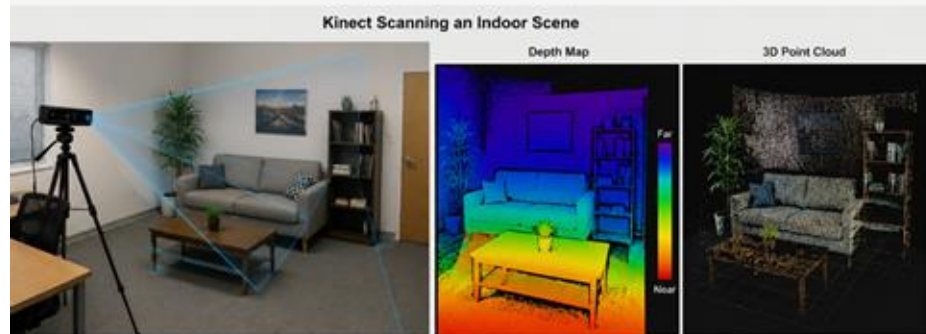
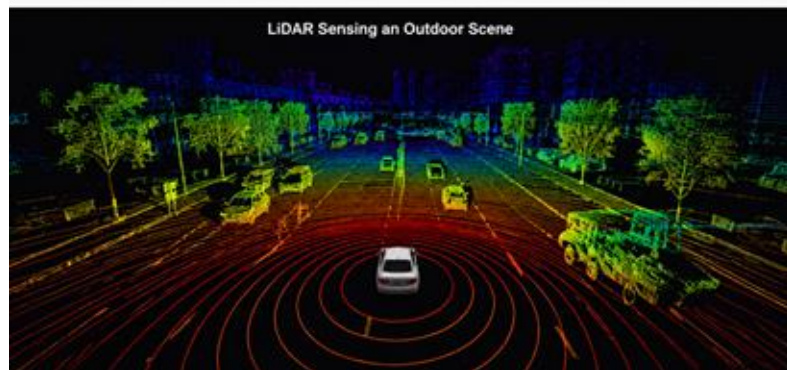
3D Representation - Explicit

Point Cloud

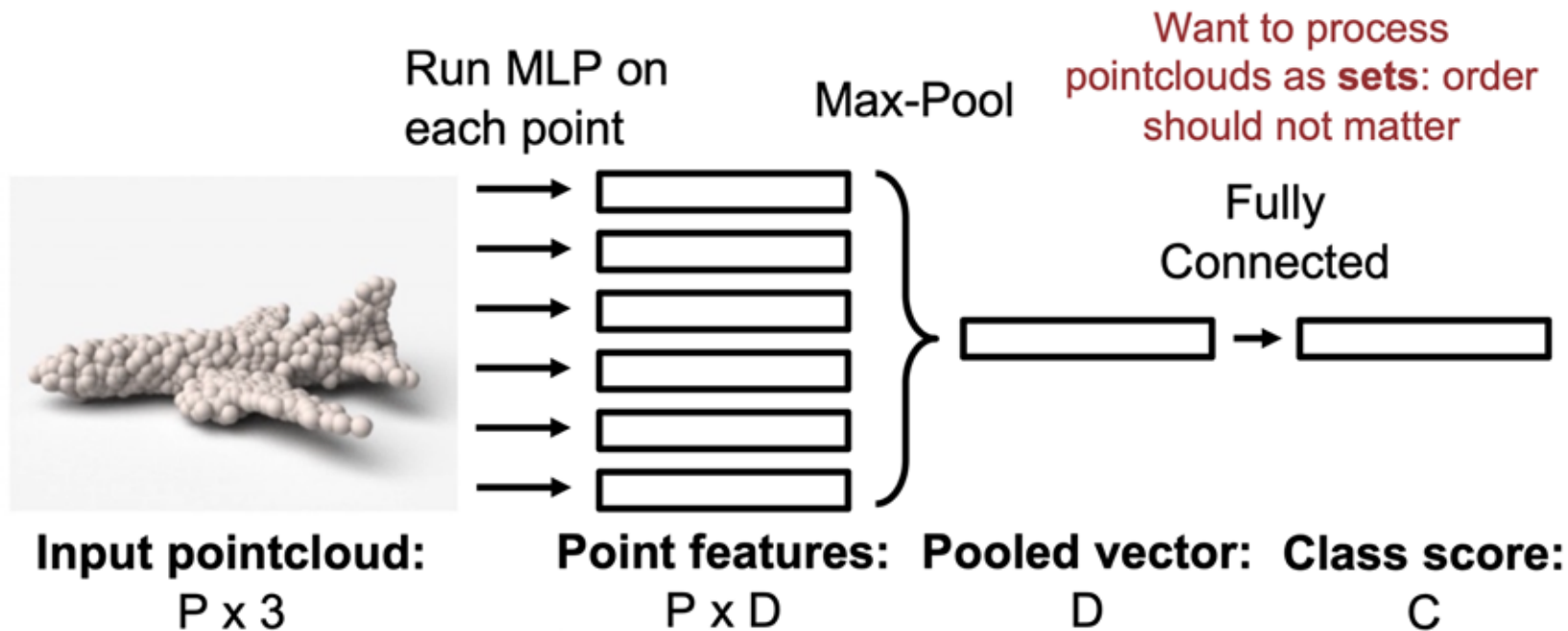
- Represented by a set of points, collection of (x, y, z) coordinates, possibly with normal
- Simple representation: **only points**, no connectivity
- (+) Can represent fine structures without huge numbers of points
- (-) Cannot easily extract the surface



3D Representation - Explicit



Processing 3D point Cloud

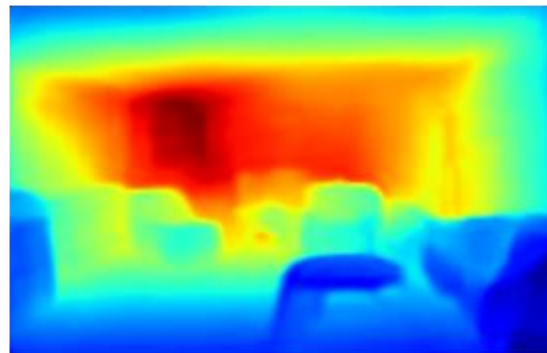


3D Representation - Explicit

- Depth Map



RGB - Image



Depth Map

3D Points: $\mathbf{X}_i = \mathbf{K}^{-1} d_i \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix}$ where $\mathbf{K}$ is the camera intrinsic matrix.

3D Representation - Explicit

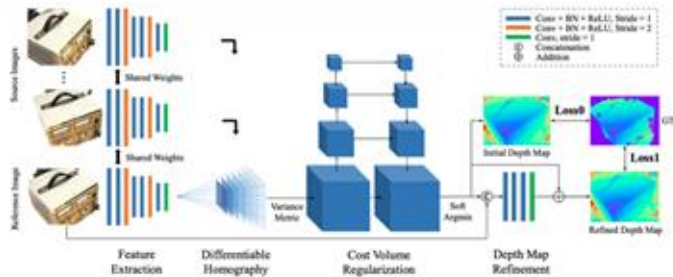


Multi-view Stereo

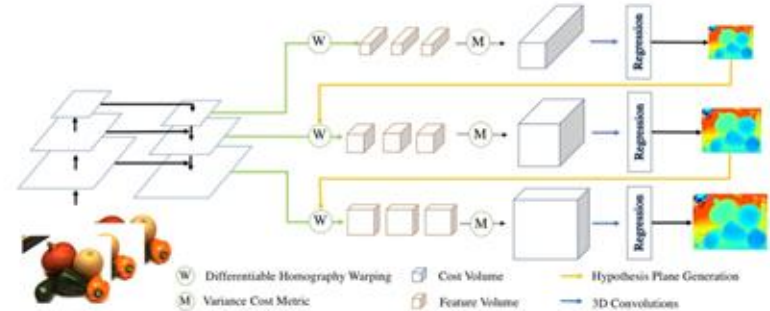


Monocular Depth Estimation, e.g. Depth Anything 3

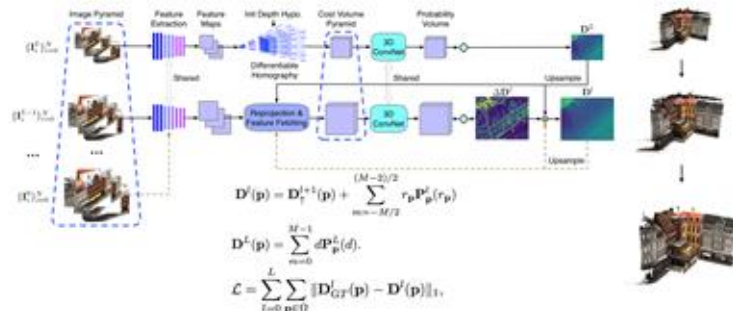
3D Representation - Explicit



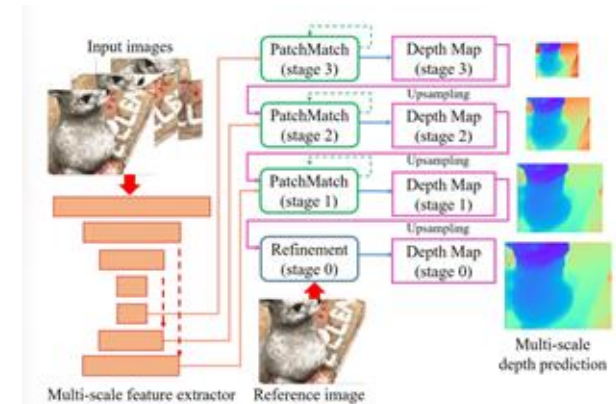
MVSNet: Depth Inference for Unstructured Multi-view Stereo, Yan et al. ECCV2018



Cascade Cost Volume for High-Resolution Multi-View Stereo and Stereo Matching, Gu et al. CVPR2020

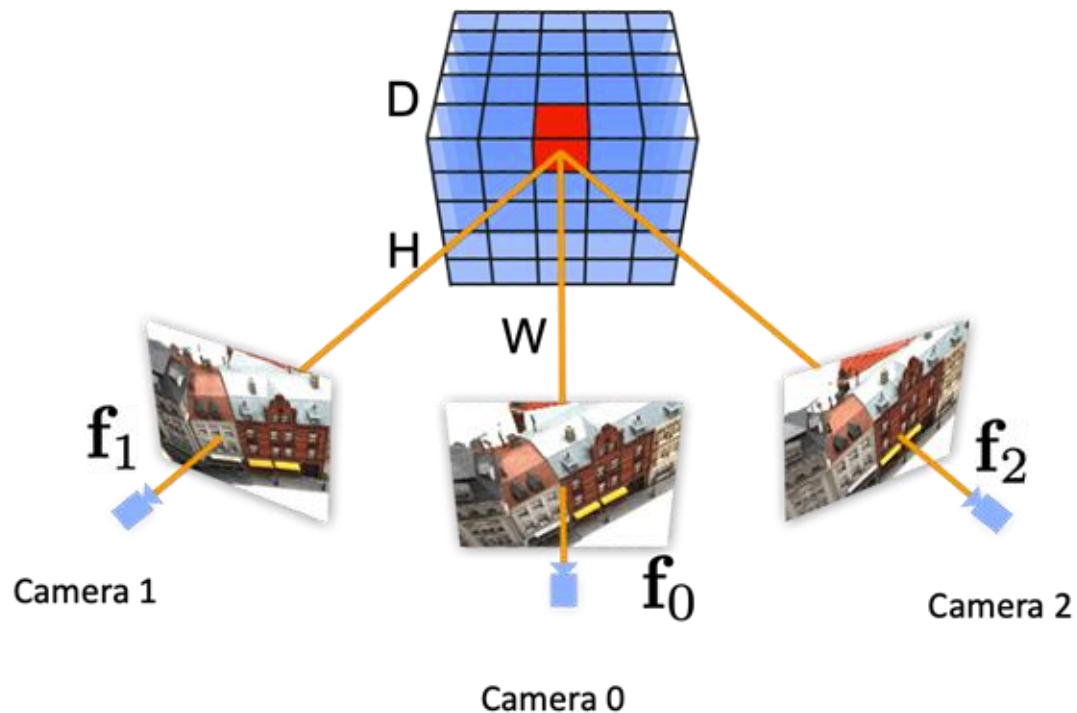


Cost Volume Pyramid Based Depth Inference for Multi-View Stereo, Yang et al. CVPR2020



PatchmatchNet: Learned Multi-View Patchmatch Stereo, Wang et al. CVPR2021

Cost Volume



$$\mathbf{c} = \frac{1}{n} \sum_{i=0}^n (\mathbf{f}_i - \bar{\mathbf{f}})^2$$

- It measures the photo-consistency across multiple views.

3D Representation - Explicit

Monocular Depth Estimation

Depth Map Prediction from a Single Image using a Multi-Scale Deep Network

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¹Dept. of Computer Science, Courant Institute, New York University

²Facebook AI Research

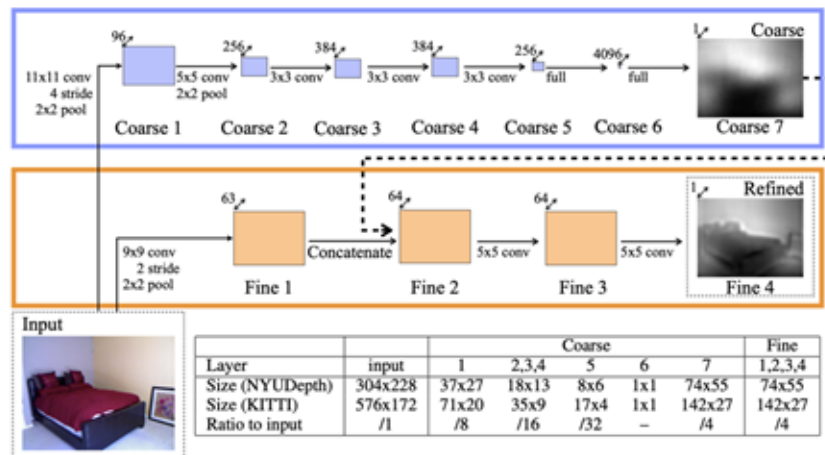


Figure 1: Model architecture.

Key idea:

1. The global scale of a scene is ambiguous.
2. Multi-scale prediction.
3. Scale-invariant training loss.

$$d_i = \log y_i - \log y_i^*$$

$$\begin{aligned} D(y, y^*) &= \frac{1}{2n^2} \sum_{i,j} ((\log y_i - \log y_j) - (\log y_i^* - \log y_j^*))^2 \\ &= \frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \sum_{i,j} d_i d_j = \frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \left(\sum_i d_i \right)^2 \end{aligned}$$

3D Representation - Explicit

Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data, Yang et al. CVPR 2024

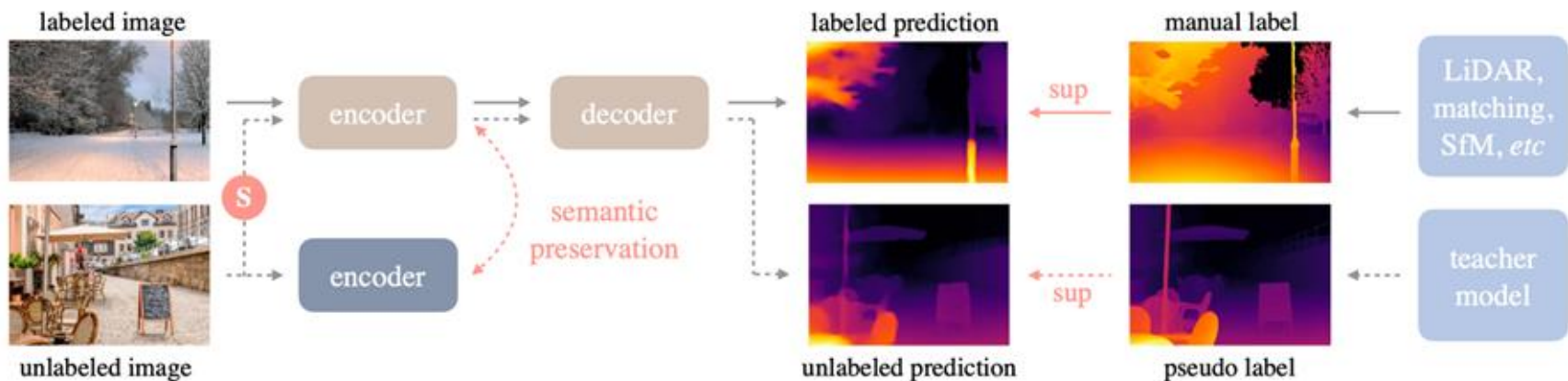


Core Idea :

- Scaling up the training data, 1.5 million labelled images + 62 million unlabelled images
- Pseudo depth maps from a pre-trained teacher network, augmented with semantic preserving loss for training.

3D Representation - Explicit

Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data, Yang et al. CVPR 2024



Core Idea :

- Scaling up the training data, 1.5 million labelled images + 62 million unlabelled images
- Pseudo depth maps from a pre-trained teacher network, Semantic preservation, DINO V2 features

3D Representation - Implicit

Signed Distance Function (SDF)

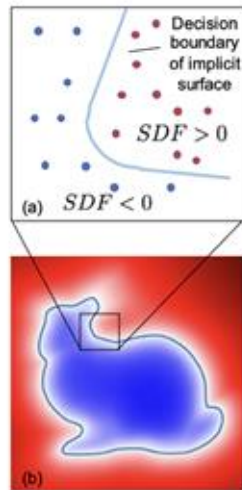
Let $\Omega \subset \mathbb{R}^n$ be a solid region with boundary $\partial\Omega$.

The signed distance function $d_\Omega : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$d_\Omega(\mathbf{x}) = \begin{cases} -\text{dist}(\mathbf{x}, \partial\Omega), & \mathbf{x} \in \Omega, \\ 0, & \mathbf{x} \in \partial\Omega, \\ \text{dist}(\mathbf{x}, \partial\Omega), & \mathbf{x} \notin \overline{\Omega}, \end{cases} \quad \mathbf{x} \in \mathbb{R}^n,$$

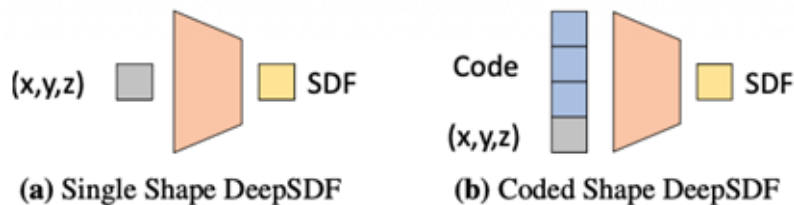
where

$$\text{dist}(\mathbf{x}, A) = \inf_{\mathbf{y} \in A} \|\mathbf{x} - \mathbf{y}\|_2, \quad A \subset \mathbb{R}^n.$$



3D Representation - Implicit

Deep SDF



DeepSDF network (MLPs) outputs SDF value at a 3D query location for a single shape only or multiple shape represented by a latent code.

$$\arg \min_{\theta, \{z_i\}_{i=1}^N} \sum_{i=1}^N \left(\sum_{j=1}^K \mathcal{L}(f_{\theta}(z_i, \mathbf{x}_j), s_j) + \frac{1}{\sigma^2} \|z_i\|_2^2 \right)$$



Figure 1: DeepSDF represents signed distance functions (SDFs) of shapes via latent code-conditioned feed-forward decoder networks. Above images are raycast renderings of DeepSDF interpolating between two shapes in the learned shape latent space. Best viewed digitally.

- Compact 3D shape representation learning
- Shape Generation (See Fig.1)
- Shape Completion (eg. 3D shape completion from partial observation, such as a depth map)

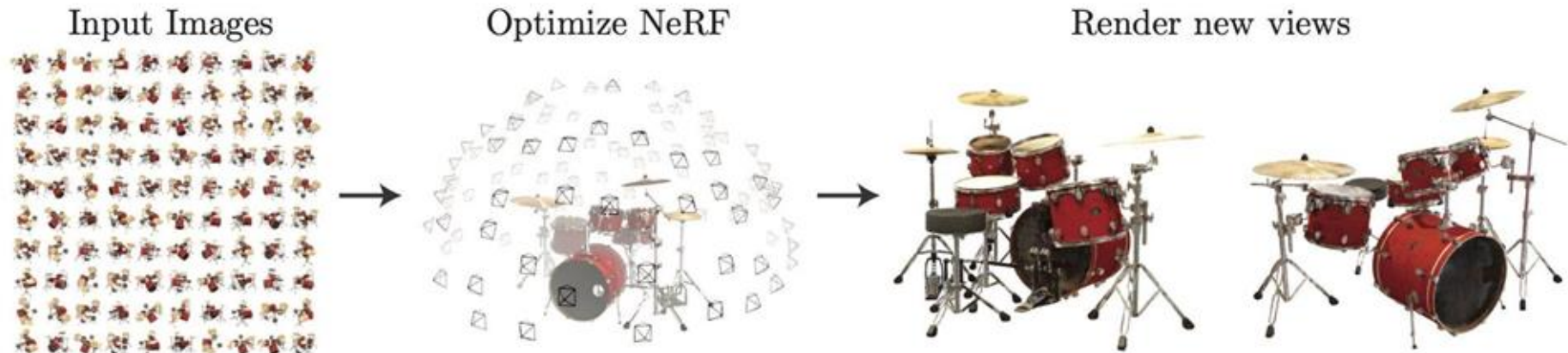
3D Representation - Implicit

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020



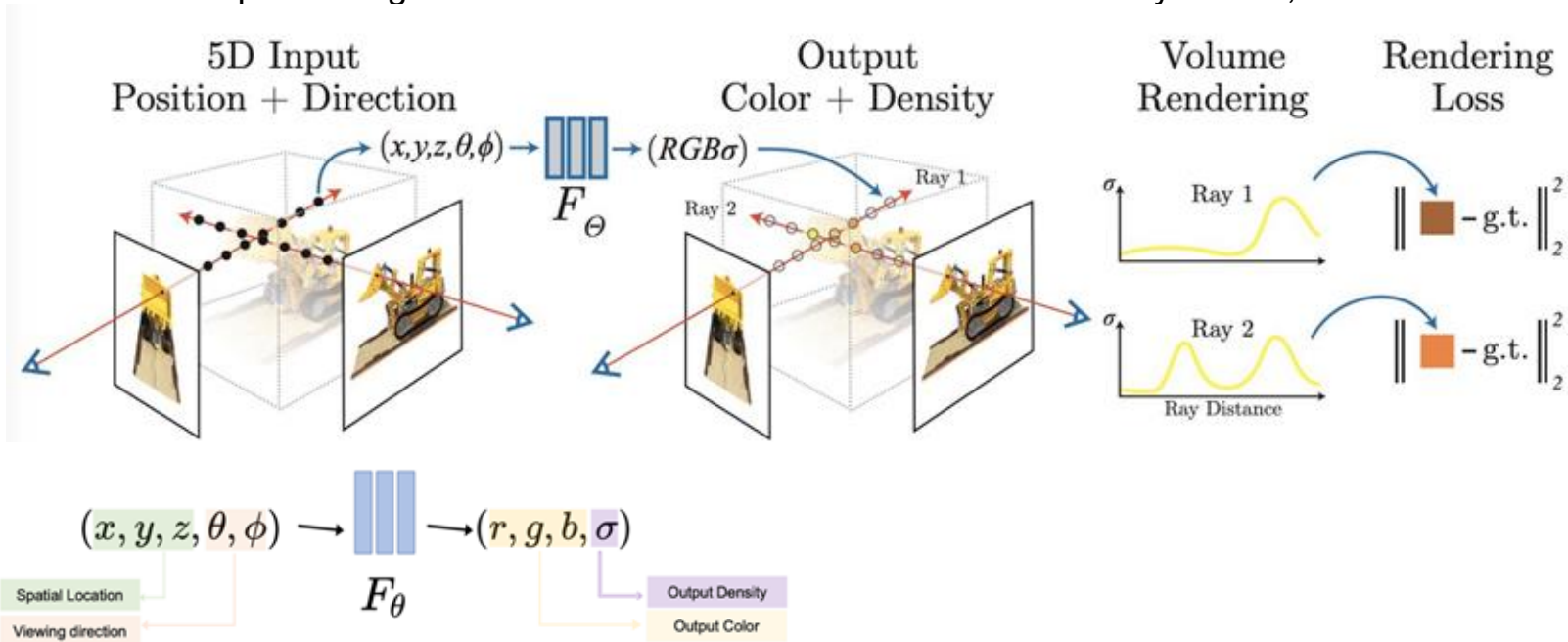
3D Representation - Implicit

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020

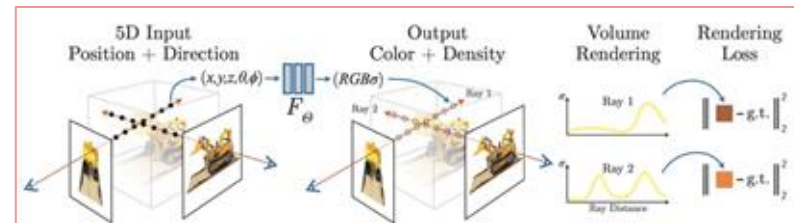


3D Representation - Implicit

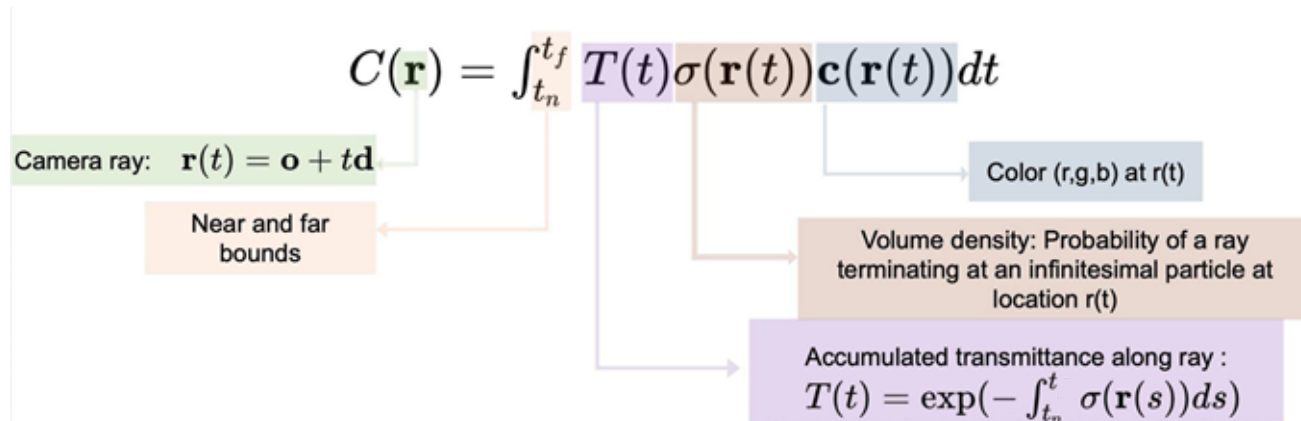
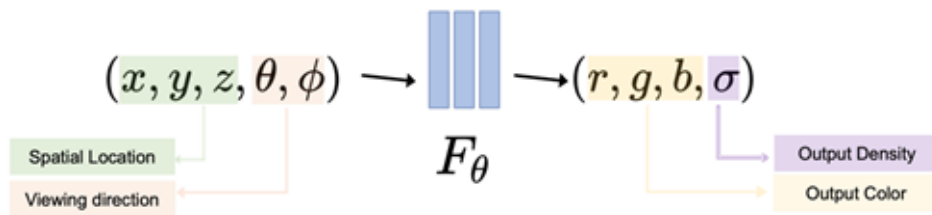
NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020



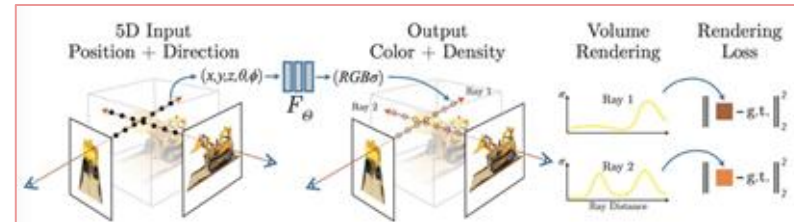
3D Representation - Implicit



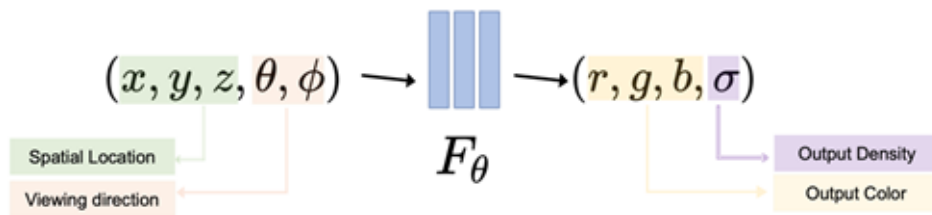
NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020



3D Representation - Implicit



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020



$$\hat{C}(\mathbf{r}) = \sum_{i=1}^N T_i (1 - \exp(-\sigma_i \delta_i)) \mathbf{c}_i$$

Uniform N samples
 $t_i \sim \mathcal{U}\left[t_n + \frac{i-1}{N}(t_f - t_n), t_n + \frac{i}{N}(t_f - t_n)\right]$

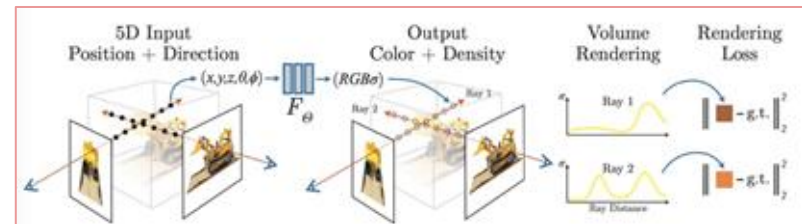
Distance between adjacent samples
 $\delta_i = t_{i+1} - t_i$

Alpha (in traditional alpha compositing)
 $\alpha_i = 1 - \exp(-\sigma_i \delta_i)$

Accumulated transmittance along ray
 $T_i = \exp\left(-\sum_{j=1}^{i-1} \sigma_j \delta_j\right)$

Color (r,g,b) at t_i

3D Representation - Implicit



NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, Mildenhall et al. ECCV 2020

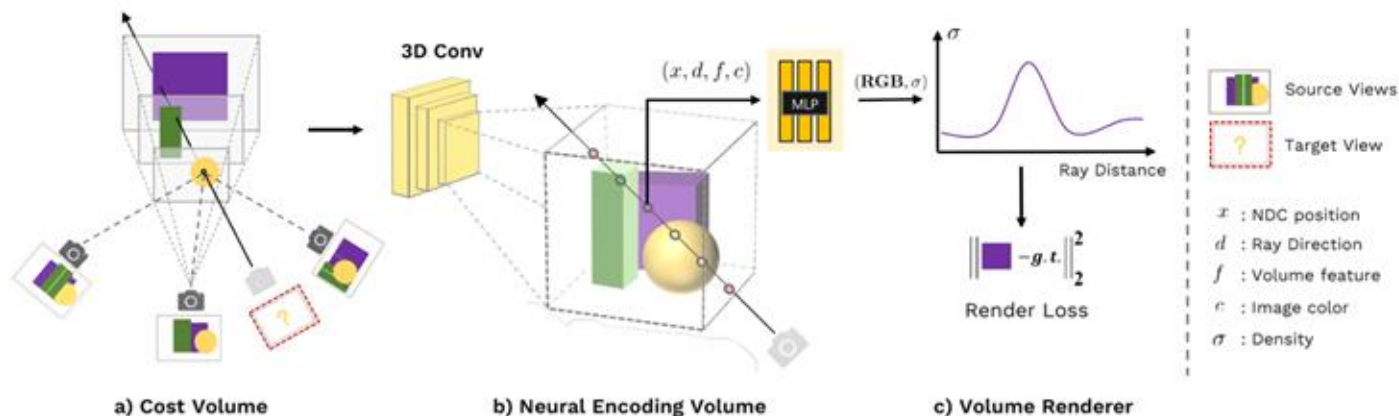
Rendering Depth:

$$\hat{D}(\mathbf{r}) = \sum_{i=1}^N T_i (1 - \exp(-\sigma_i \delta_i)) d_i$$



3D Representation - Implicit

MVSNeRF: Fast Generalizable Radiance Field Reconstruction from Multi-View Stereo, Chen et al. ICCV2021



$$\sigma, r = \text{MVSNeRF}(x, d; I_i, \Phi_i)$$

Not per-scene optimisation. Generalisable across datasets.

Key Idea: feature from pre-trained cost volume for color and density estimation

3D Representation - Implicit

MVSNeRF: Fast Generalizable Radiance Field Reconstruction from Multi-View Stereo, Chen et al. ICCV21



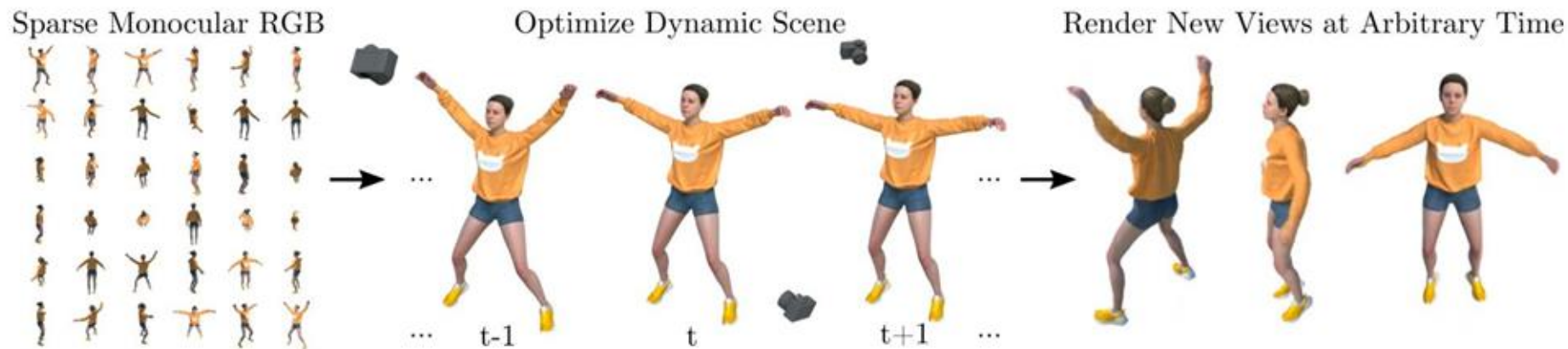
$$\sigma, r = \text{MVSNeRF}(x, d; I_i, \Phi_i)$$

Generalisation: MVS-NeRF + 6 min fine-tuning vs. NeRF: 5.1 h per-scene optimisation
— approximately 51× faster.

3D Representation - Implicit

Modelling Dynamic Object

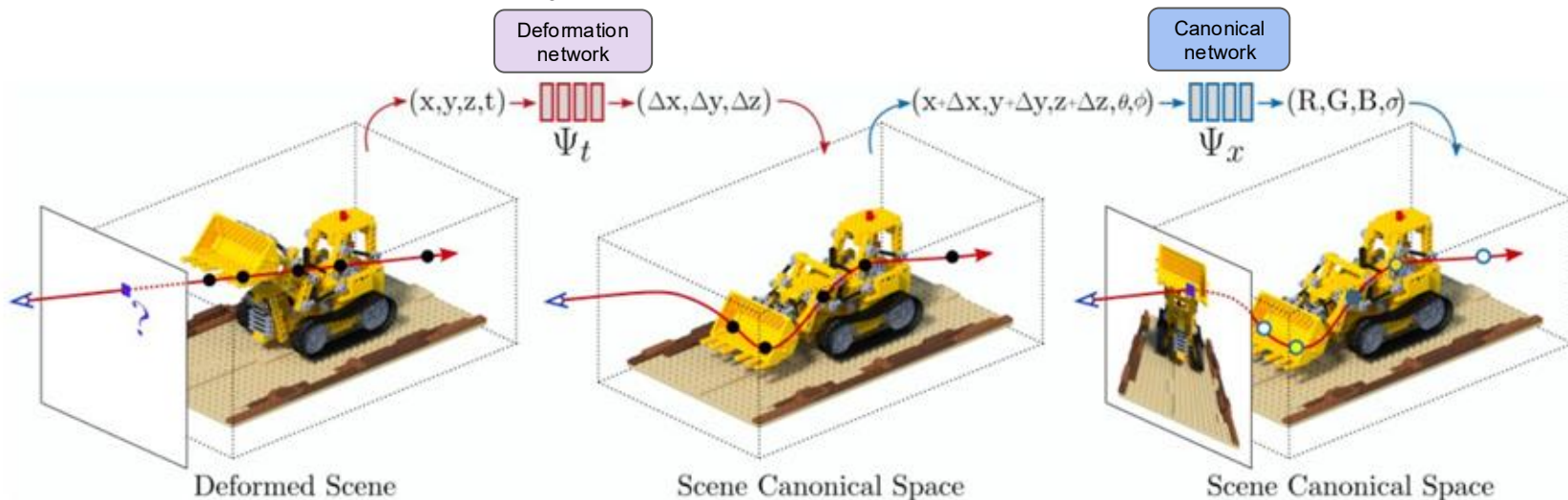
D-NeRF: Neural Radiance Fields for Dynamic Scenes, Pumarola et al. CVPR 2021



3D Representation - Implicit

Modelling Dynamic Object

D-NeRF: Neural Radiance Fields for Dynamic Scenes, Pumarola et al. CVPR 2021



Representing Dynamic object/scene

3D Representation - Explicit

3D Gaussian Splatting for Real-Time Radiance Field Rendering, Kerbl, et al. Siggraph 2023



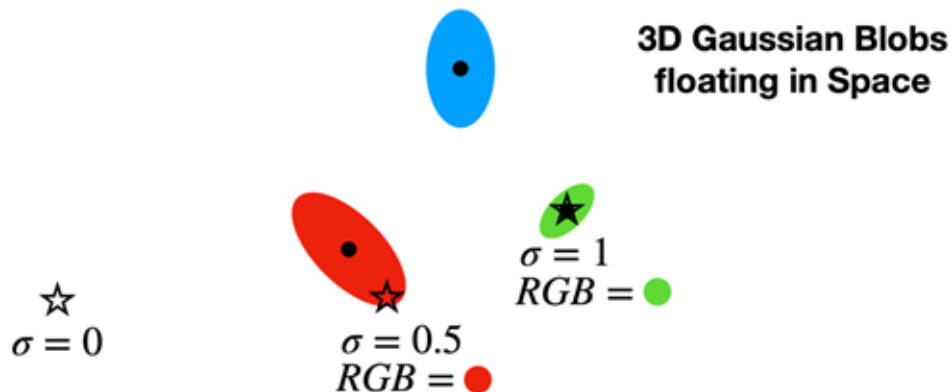
Rendering time:

- 30~100+ FPS (GS)
- several seconds (NeRF)

3D Representation - Explicit

Key Idea

- Representing the whole scene as Gaussian primitives (blobs)
- Optimisation for Parameterized Radiance Field sparsely, only where density is nonzero
- Gaussian center, Pose, Spherical harmonics parameters, Scale, Density

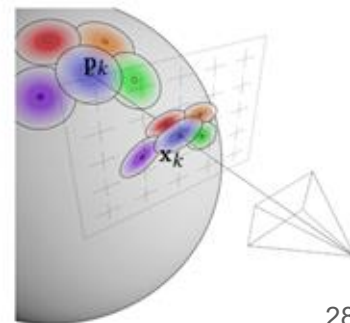
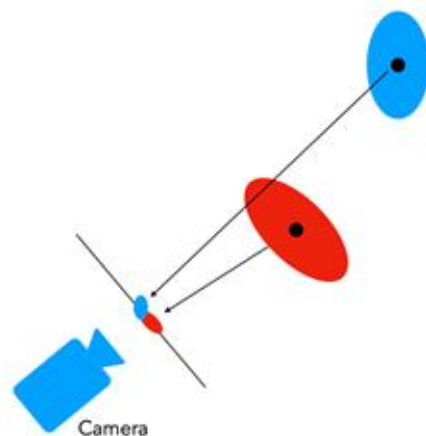
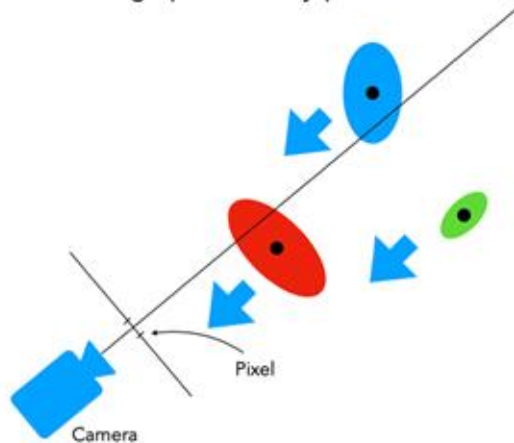


3D Representation - Explicit

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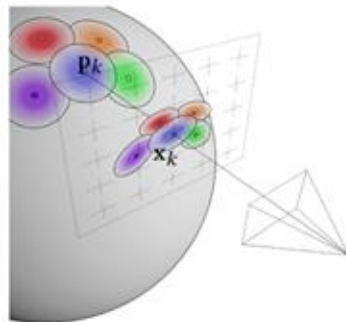
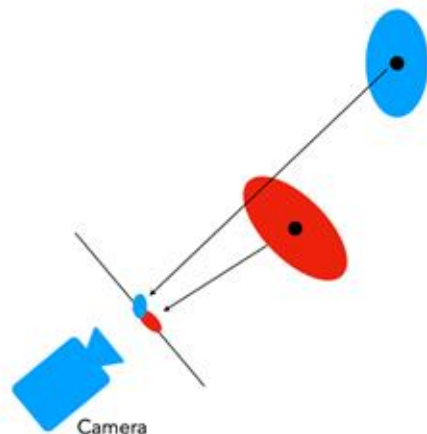
Project onto image primitive by primitive



3D Representation - Explicit

Key Idea

- Representing the whole scene as Gaussian primitives (blobs)
- Optimisation for Parameterized Radiance Field sparsely, only where density is nonzero
- Gaussian center, Pose, Spherical harmonics parameters, Scale, Density



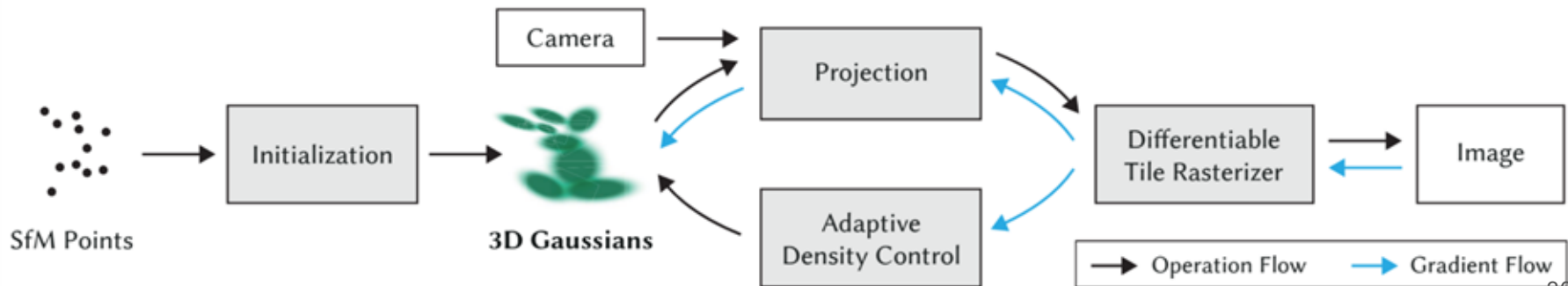
$$C = \sum_{i \in \mathcal{N}} c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad (3)$$

where c_i is the color of each point and α_i is given by evaluating a 2D Gaussian with covariance Σ [Yifan et al. 2019] multiplied with a learned per-point opacity.

3D Representation - Explicit

Key Idea

- Representing the whole scene as Gaussian primitives (blobs)
- Optimisation for Parameterized Radiance Field sparsely, only where density is nonzero
- Gaussian center, Pose, Spherical harmonics parameters, Scale, Density
- Initialise Gaussians from sfm points to avoid local minimum



3D Representation - Explicit

Dynamic Scene

Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis, Luiten et al. 3DV 2024



3D Representation - Explicit

Dynamic Scene

Dynamic 3D Gaussians: Tracking by Persistent Dynamic View Synthesis, Luiten et al. 3DV 2024

Dynamic 3D Gaussians. Our dynamic scene representation ($\mathcal{S}$) is parameterized by a set of Dynamic 3D Gaussians, each of which has the following parameters:

- 1) a 3D center for each timestep (x_t, y_t, z_t).
- 2) a 3D rotation for each timestep parameterized by a quaternion (qw_t, qx_t, qy_t, qz_t).
- 3) a 3D size in standard deviations (consistent over all timesteps) (sx, sy, sz)
- 4) a color (consistent over all timesteps) (r, g, b)
- 5) an opacity logit (consistent over all timesteps) (o)
- 6) a background logit (consistent over timesteps) (bg)

This gives a total of $7t + 8$ parameters for each Gaussian.

$$C_{\text{pix}} = \sum_{i \in \mathcal{S}} c_i f_{i,\text{pix}}^{2\text{D}} \prod_{j=1}^{i-1} (1 - f_{j,\text{pix}}^{2\text{D}})$$

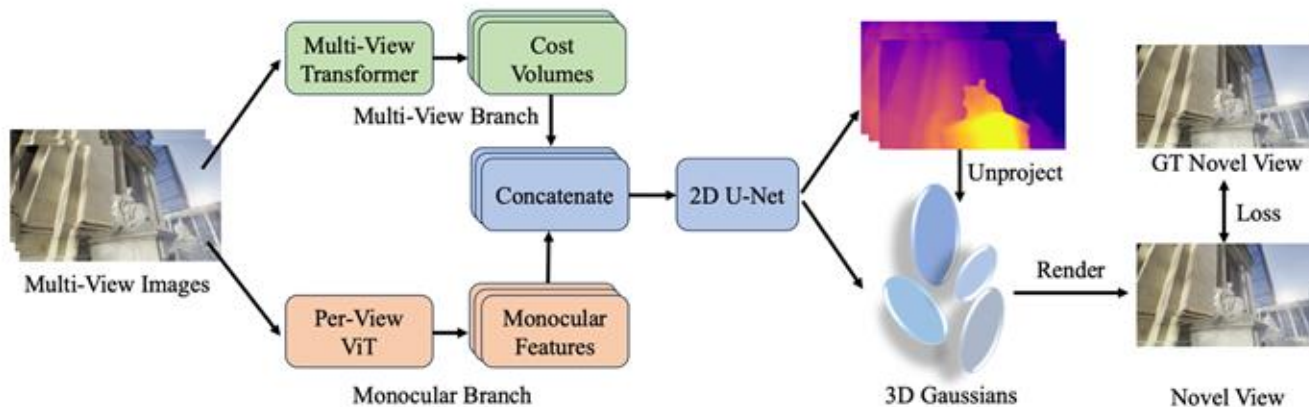
$$\mathcal{L}_{i,j}^{\text{rigid}} = w_{i,j} \|(\mu_{j,t-1} - \mu_{i,t-1}) - R_{i,t-1} R_{i,t}^{-1} (\mu_{j,t} - \mu_{i,t})\|_2$$

$$\mathcal{L}^{\text{rot}} = \frac{1}{k|\mathcal{S}|} \sum_{i \in \mathcal{S}} \sum_{j \in \text{knn}_{i;k}} w_{i,j} \|\hat{q}_{j,t} \hat{q}_{j,t-1}^{-1} - \hat{q}_{i,t} \hat{q}_{i,t-1}^{-1}\|_2$$

3D Representation - explicit

Feed forward Gaussian Splatting

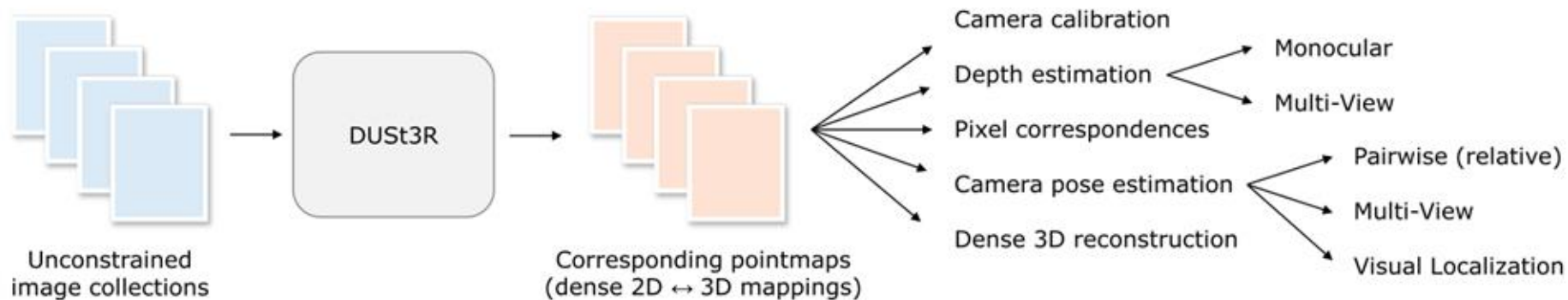
DepthSplat: Connecting Gaussian Splatting and Depth, Xu et al. CVPR 2025



- Pixelwise Gaussian,
- Predicted depth as the Gaussian center
- MVS+Monocular feature, depth, image for gaussian covariance, opacity and color predictions.
- Losses: Depth estimation loss and image synthesis loss

3D Foundation Model

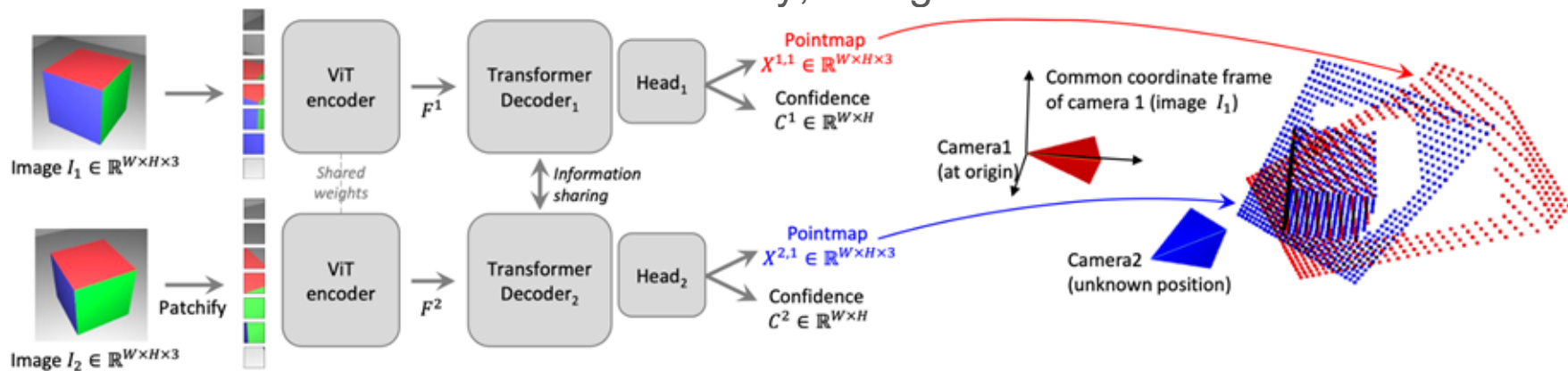
DUSt3R: Geometric 3D Vision Made Easy, Wang et al. CVPR 2024



Goal: A pre-trained model for multiple 3D vision tasks.

3D Foundation Model

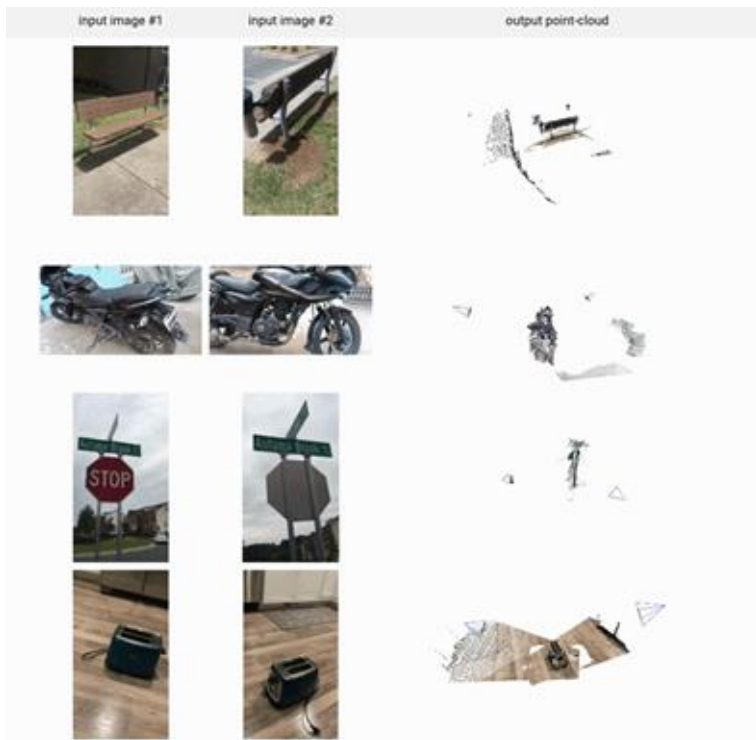
DUST3R: Geometric 3D Vision Made Easy, Wang et al. CVPR 2024



Key Idea:

- Input: a pair of images, output: two point cloud (in coordinate. system by I_1), two confidence maps
- Directly predict point cloud, not depth maps
- Pre-trained on a large amount of data
- Obtain pixel matches, relative and absolute cameras
- Training on a large amount of data (8.5M pairs)

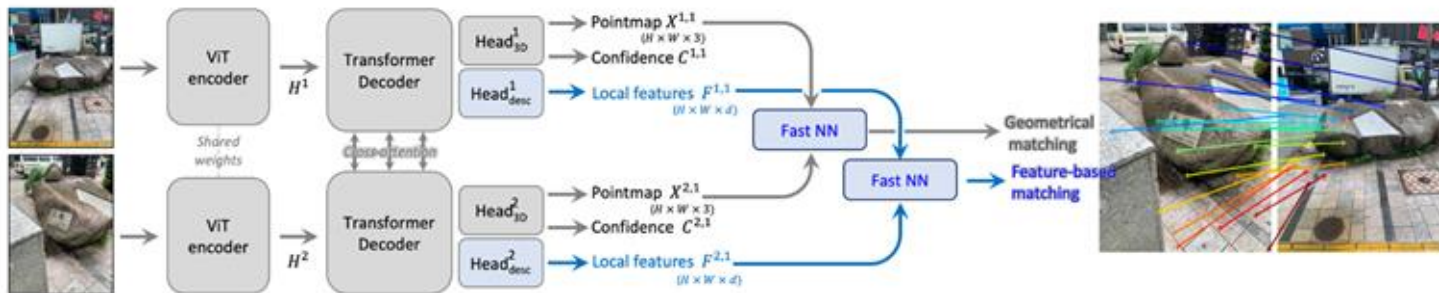
3D Foundation Model



- End-to-end 3D reconstruction pipeline from un-calibrated and un-posed images
- Based on a generic transformer architecture,
- Not enforcing any geometric constraints at inference, but being able to benefit from powerful pre-training schemes

3D Foundation Model

Grounding Image Matching in 3D with MAST3R, Leroy et al, ECCV24



Key Idea:

- Comparing with DUST3R, Introducing 2D and 3D matching
- leverage additional infoNCE loss for matching given ground truth matching info during training

$$\mathcal{L}_{\text{match}} = - \sum_{(i,j) \in \hat{\mathcal{M}}} \log \frac{s_{\tau}(i,j)}{\sum_{k \in \mathcal{P}^1} s_{\tau}(k,j)} + \log \frac{s_{\tau}(i,j)}{\sum_{k \in \mathcal{P}^2} s_{\tau}(i,k)},$$

with $s_{\tau}(i,j) = \exp [-\tau D_i^{1\top} D_j^2]$.

3D Foundation Model

MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors, CVPR25



Uncalibrated RGB
8x Playback



3D Foundation Model

MASt3R-SLAM: Real-Time Dense SLAM with 3D Reconstruction Priors, CVPR25



Key Idea:

- 1) Leverage the output from MASt3R as prior
- 2) Resolve scale inconsistency
- 3) Fast tracking
- 4) relocalisation

3D Foundation Model

VGGT: Visual Geometry Grounded Transformer, Wang et al. CVPR 2025



Figure 1. **VGGT** is a large feed-forward transformer with minimal 3D-inductive biases trained on a trove of 3D-annotated data. It accepts up to hundreds of images and predicts cameras, point maps, depth maps, and point tracks for all images at once in less than a second, which often outperforms optimization-based alternatives without further processing.

3D Foundation Model

VGGT: Visual Geometry Grounded Transformer, Wang et al. CVPR 2025

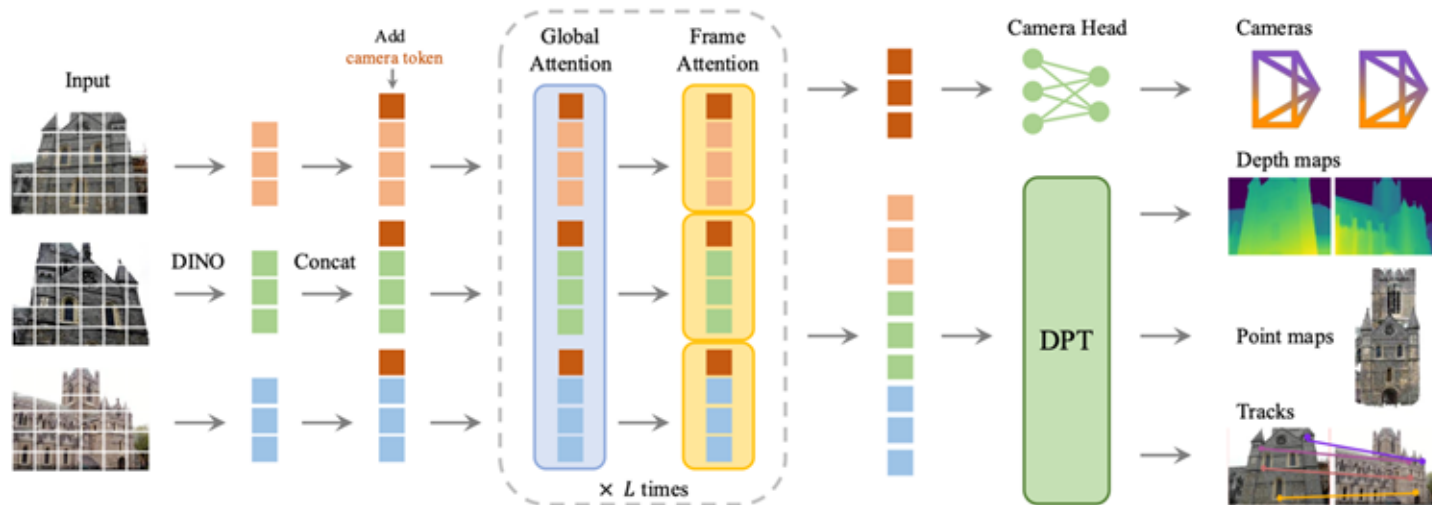


Figure 2. **Architecture Overview.** Our model first patchifies the input images into tokens by DINO, and appends camera tokens for camera prediction. It then alternates between frame-wise and global self attention layers. A camera head makes the final prediction for camera extrinsics and intrinsics, and a DPT [87] head for any dense output.

3D Foundation Model

VGGT: Visual Geometry Grounded Transformer, Wang et al. CVPR 2025

32 Views



More results: <https://vgg-t.github.io/>

3D Foundation Model

VGGT-SLAM: Dense RGB SLAM Optimized on the SL(4) Manifold, Maggio et al. NeurIPS 2025



RECONSTRUCTION OF LARGE BARN

3D Foundation Model

VGGT-SLAM: Dense RGB SLAM Optimized on the $SL(4)$ Manifold, Maggio et al. NeurIPS 2025



RECONSTRUCTION OF LARGE BARN

Key problem:

- 1) Allow the handling arbitrary number of frames.
- 1) Resolving the homography in $SL(4)$, (special linear) group to connect submaps.

3D Foundation Model

More on Foundation Model

- VGGT-Omega, CVPR 26,
- VGGT-Long, ICRA 26 [scalable 3D reconstruction]
- Anysplat, Siggraph 25 [Feed-forward 3D Gaussian Splatting, distill from VGGT]
- D4RT, CVPR 26 Best paper award [preliminary results for handling dynamic scenes]